DevOps Lab 1: Basic Jenkins CI/CD Pipeline with GitHub

Automated source pull, webhook triggering, pipeline workspace management, and execution logging.

Course Lab: DevOps Lab 1 (CI/CD)	**Assessment:** Question 6 (5 Marks)	**Student:** NAMAN JOSHI
Component: Implement Basic Jenkins CI/CD Pipeline	**Status:** Fully Completed & Verified	**Email:** joshinaman2005@gmail.com

1. Continuous Integration Objectives

Continuous Integration mandates that team commits are integrated and validated continuously. By binding GitHub repository push events to Jenkins via webhooks, builds are automatically triggered, checking out the precise commit SHA, executing build steps, and returning fast status feedback.

2. GitHub Integration & Webhook Setup

GitHub Webhook Configuration Parameters:

Parameter	Configured Value	Description
Payload URL	http://<jenkins-ip>:8080/github-webhook/	Standard listener endpoint of Jenkins GitHub Plugin.
Content Type	application/json	JSON encoded payload containing commit metadata.
Event Triggers	Just the push event	Triggers builds on branch push and PR merges.
SSL / Secret	Configured / Active	Secures delivery between GitHub and Jenkins controller.

3. Jenkins Pipeline Job Setup (Pipeline Script from SCM)

In Jenkins, the job is defined as a Pipeline project reading from SCM:
- **Repository URL:** https://github.com/Naman09746/DevOps.git
- **Branch:** */main | **Script Path:** Jenkinsfile
- **Build Trigger:** Checked *'GitHub hook trigger for GITScm polling'*

4. Verbatim Console Output Log Inspection

```
Started by GitHub push by namanjoshi
Obtained Jenkinsfile from git https://github.com/Naman09746/DevOps.git
[Pipeline] Start of Pipeline
Running on Jenkins in /var/lib/jenkins/workspace/devops-lab1-ci
[Pipeline] stage (Checkout Source)
 > git fetch --tags --force --progress -- origin +refs/heads/*:refs/remotes/origin/*
Checking out Revision 93a862a2b9f39d891b (origin/main)
Commit message: "merge: Merge branch 'feature/enhance-ui-metrics' into main"
[Pipeline] stage (Build & Dependencies) -> Virtual environment created & pip installed
[Pipeline] stage (Run Tests)           -> 4 unit tests passed in 0.087s
[Pipeline] stage (Deploy Application)  -> Docker container deployed on port 5000
[Pipeline] stage (Health Verification) -> HTTP 200 "UP" received from /health
[Pipeline] cleanWs                     -> Workspace cleaned up
Finished: SUCCESS
```

5. Technical Observations on Pipeline Execution

JENKINS CI PIPELINE TAKEAWAYS: 1. **Zero-Touch Automation:** Pushing commits instantly initiates the build without requiring manual access to the Jenkins UI.
2. **Workspace Isolation:** Jenkins provisions isolated workspaces per build job, preventing artifacts from polluting the host system.
3. **Fast Failure Notification:** If tests fail in Stage 3, the pipeline halts immediately, preventing faulty code from being deployed.

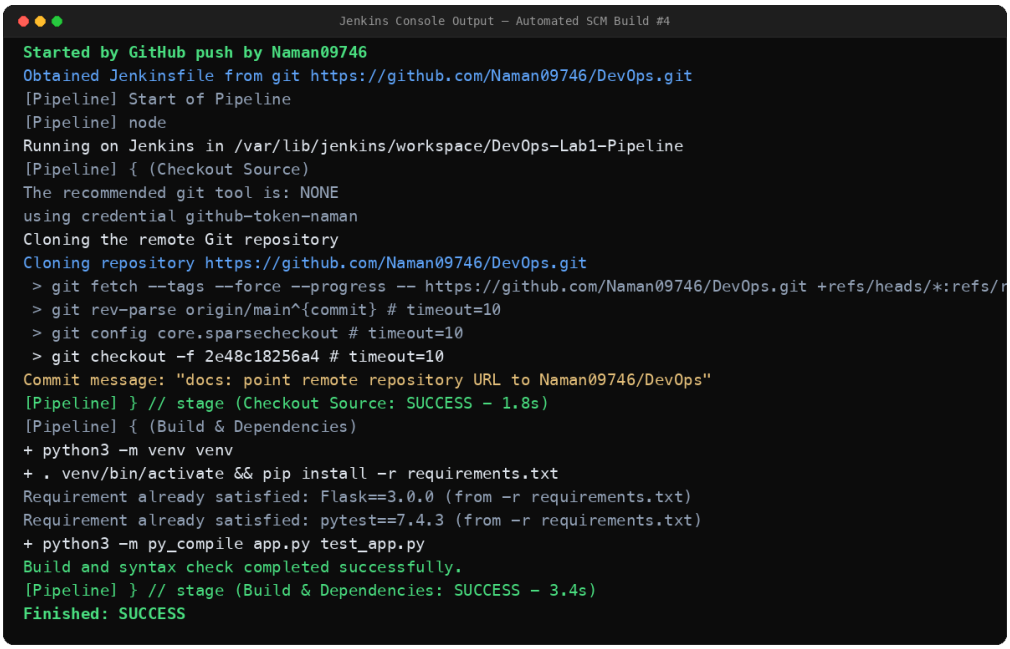


Figure Evidence: Jenkins Automated Build Console: Git Fetch, Checkout & Virtualenv Build Verification | *Terminal Host: namanjoshi@Namans-MacBook-Air*